Vehicle Stability and Control

Assignment 1

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Problem A

1.

a)

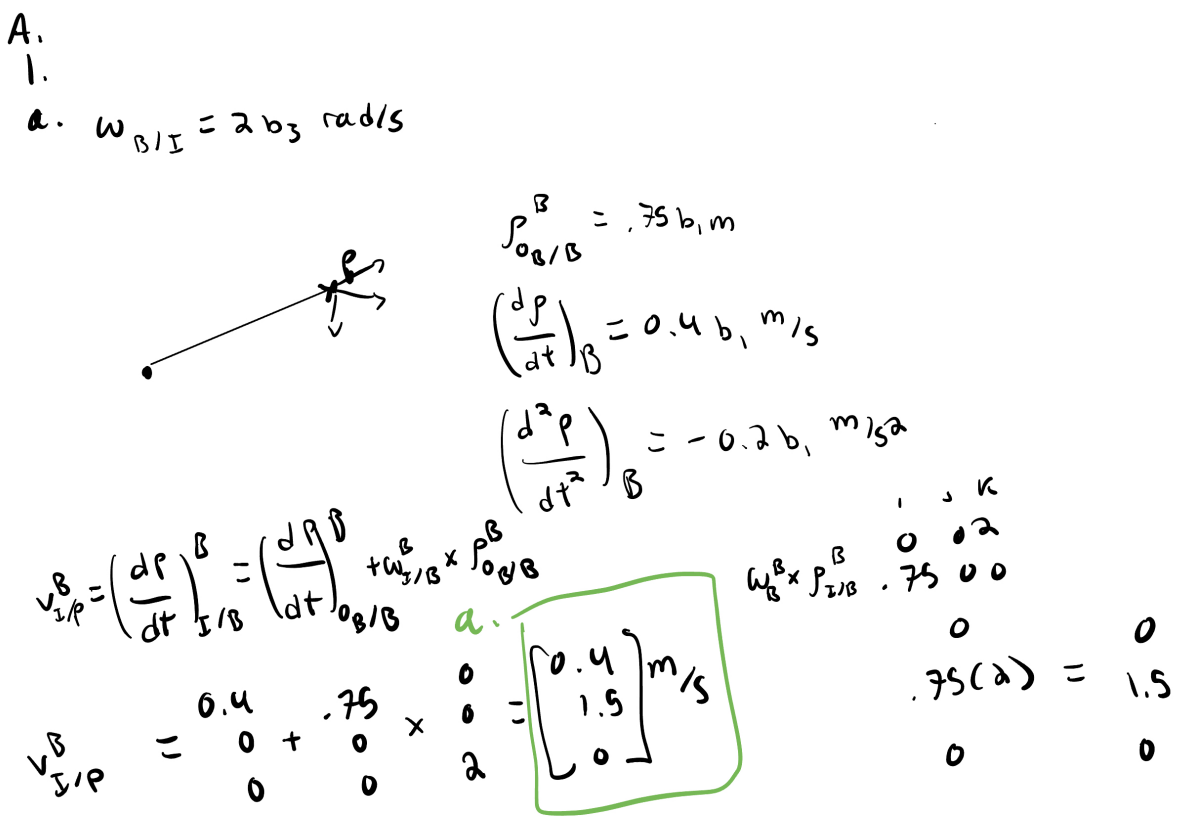


Figure 1: inertial velocity v_P of the point

Its vector components are

$$\begin{bmatrix} 0.4 \\ 1.5 \\ 0 \end{bmatrix} \tag{1}$$

Its magnitude is

$$\sqrt{0.4^2 + 1.5^2} = 1.55 \tag{2}$$

b)

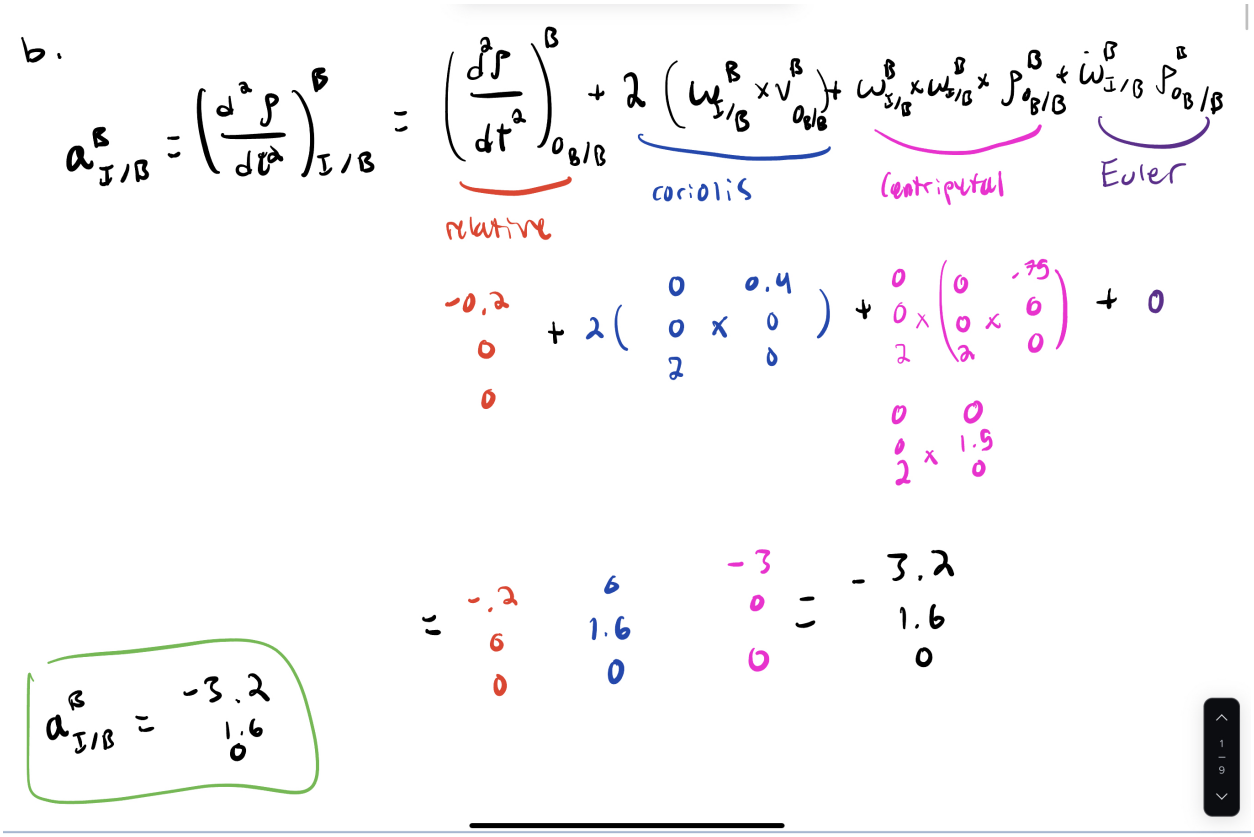


Figure 2: inertial acceleration a_P

c)

Relative acceleration is due to the increase in linear velocity of a body relative to a frame.

Coriolis acceleration is the result of the sum of two effects, but is in each case due to moving further out from a rotating frame. It is in the b2 direction, $\perp$ to the axis of rotation and in the direction of rotation (bodies speed up the greater their distance grows from the center of rotation.

Centripetal acceleration is the result of a Newtonian force balance that "holds" a stationary (wrt the rotating frame) body in its rotating reference frame without other external forces.

Euler acceleration is the result of increasing angular velocity of the rotating frame, which speeds up quantities "fixed" in its frame proportionally to their perpendicular distance. It is zero in this case.

d)

All terms aside from the relative acceleration would vanish, as they depend on $\rho_{O_b/B}$, which is 0. The inertial acceleration would not be zero due to the value of $\left(\frac{d^2P}{dt^2}\right)_B^B$

Problem B

1.

a)

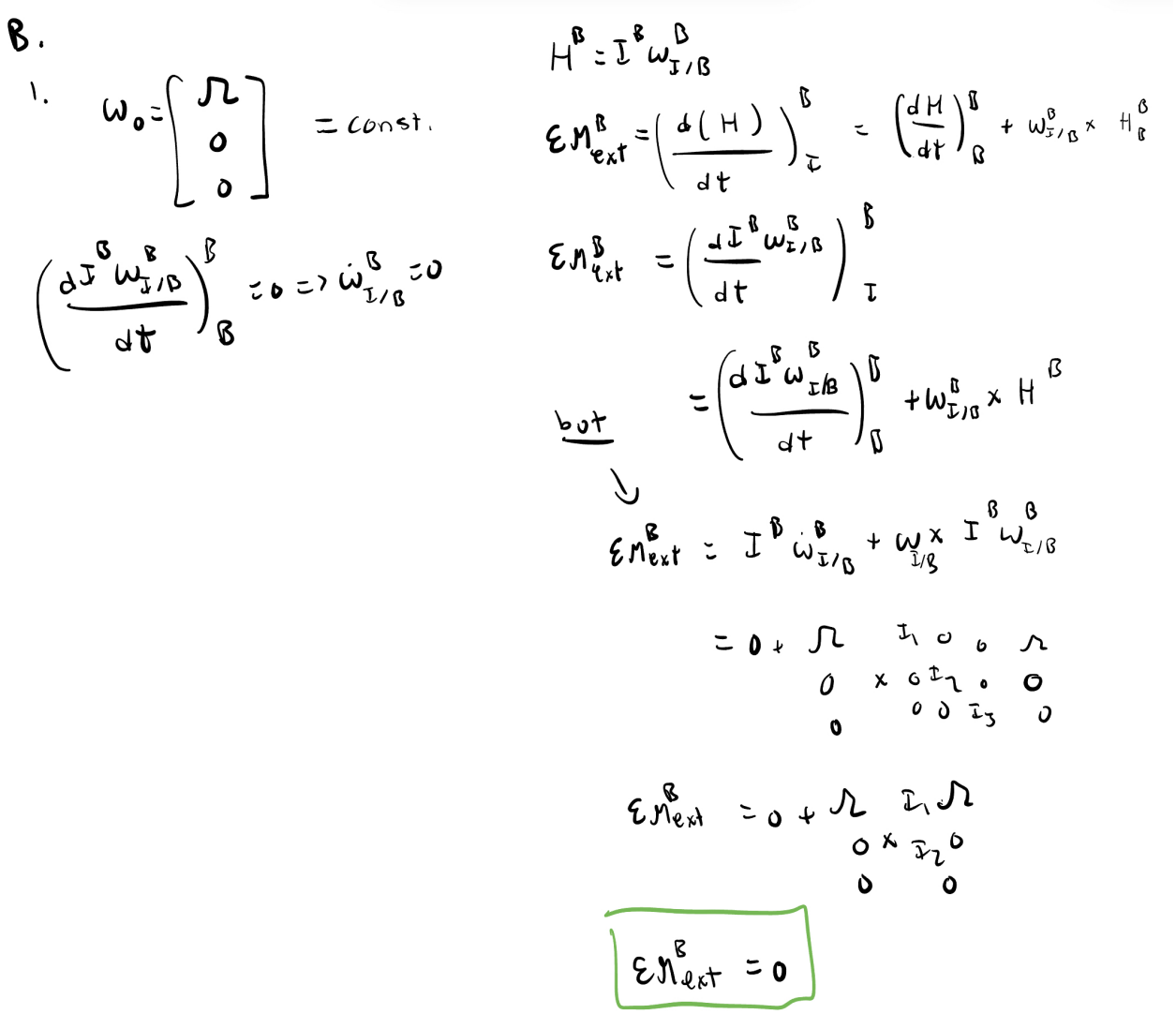


Figure 3: Torque-free motion verification

This is only a statement about the acceleration of the bodies angular velocity. It still rotates and changes orientation with respect to an inertial frame

$$\omega^B_{I/B} \neq 0 \tag{3}$$

For any point P on the body, an inertial observer will see the velocity

$$v_P = \omega \times \rho_P \tag{4}$$

b)

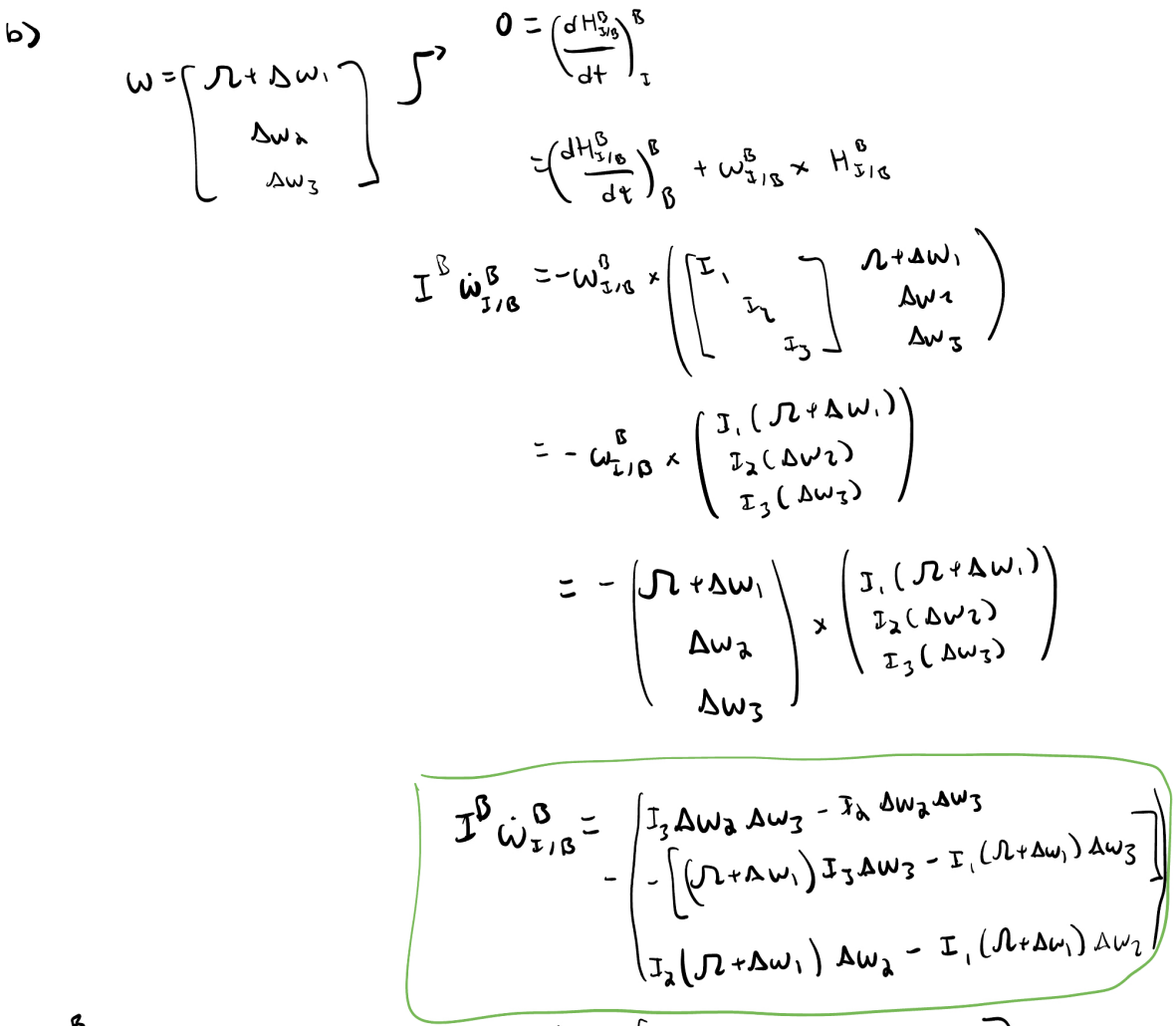


Figure 4: Exact perturbation equations

c)

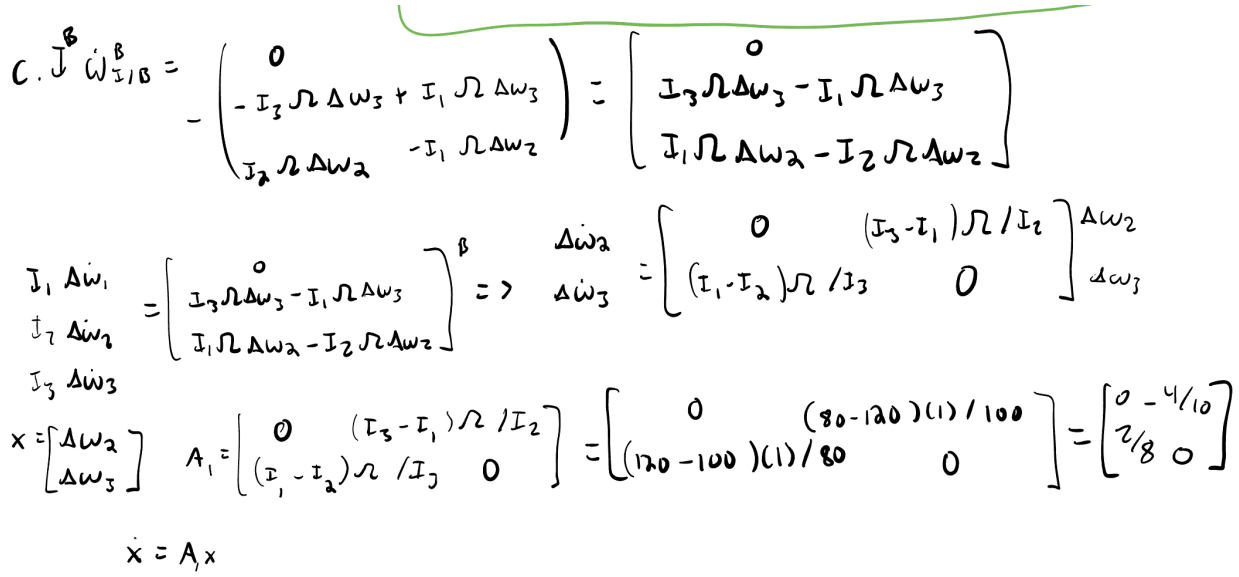


Figure 5: Symbolic and numerical expressions for A

d)

The trim condition is ω_0, the perturbation is $\Delta\omega$. so

$$x = x_0 + \Delta x \Rightarrow \omega = \omega_0 + \Delta\omega \tag{5}$$

we have an expression for the system dynamics

$$\dot{\Delta x} = f(x) \tag{6}$$

which from Euler's equation is

$$\omega^B_{I/B} = f(\omega_0 + \Delta\omega) \tag{7}$$

we can neglect the $\mathcal{O}(\Delta) \geqslant 2$ in $f(\omega_0 + \Delta\omega$ and arrive at a linear system

$$\dot{\Delta x} = A\Delta x \tag{8}$$

Problem C

1.

a)

C. Natural Modes by hand

$A_1 = \begin{bmatrix} 0 & -4/10 \\ 2/8 & 0 \end{bmatrix}$, $\det(\lambda I - A) = 0$ $\det\begin{vmatrix} \lambda & +4/10 \\ -2/8 & \lambda \end{vmatrix} = 0$

$= \lambda^2 - 4/10(-2/8)$

$= \lambda^2 + 1/10 = 0 \Rightarrow \lambda = \pm\sqrt{1/10}\, i$

$e^{\pm j\frac{1}{\sqrt{10}}t}$

$\lambda = \sigma \pm \omega_n$

$\omega_n = 1/\sqrt{10}$

Figure 6: Characteristic equation and two eigenvalues by hand

b)

The resulting natural motion is neutrally stable, oscillatory, and has no finite decay time constant.

c)

Complex conj. pair results in soln $e^{\pm j\omega_n t} = \cos|\lambda| t \pm i\sin|\lambda| t$

C. $\lambda = \sigma \pm j\omega_n$

$\omega_n = 1/\sqrt{10}$

$f = 2\pi\omega_n = 2\pi/\sqrt{10}$

$T = 1/f = \sqrt{10}/2\pi$

↓

Re $\Rightarrow$ oscillates with constant amplitude

Figure 7: Oscillatory frequency, cycle frequency, and period

d)

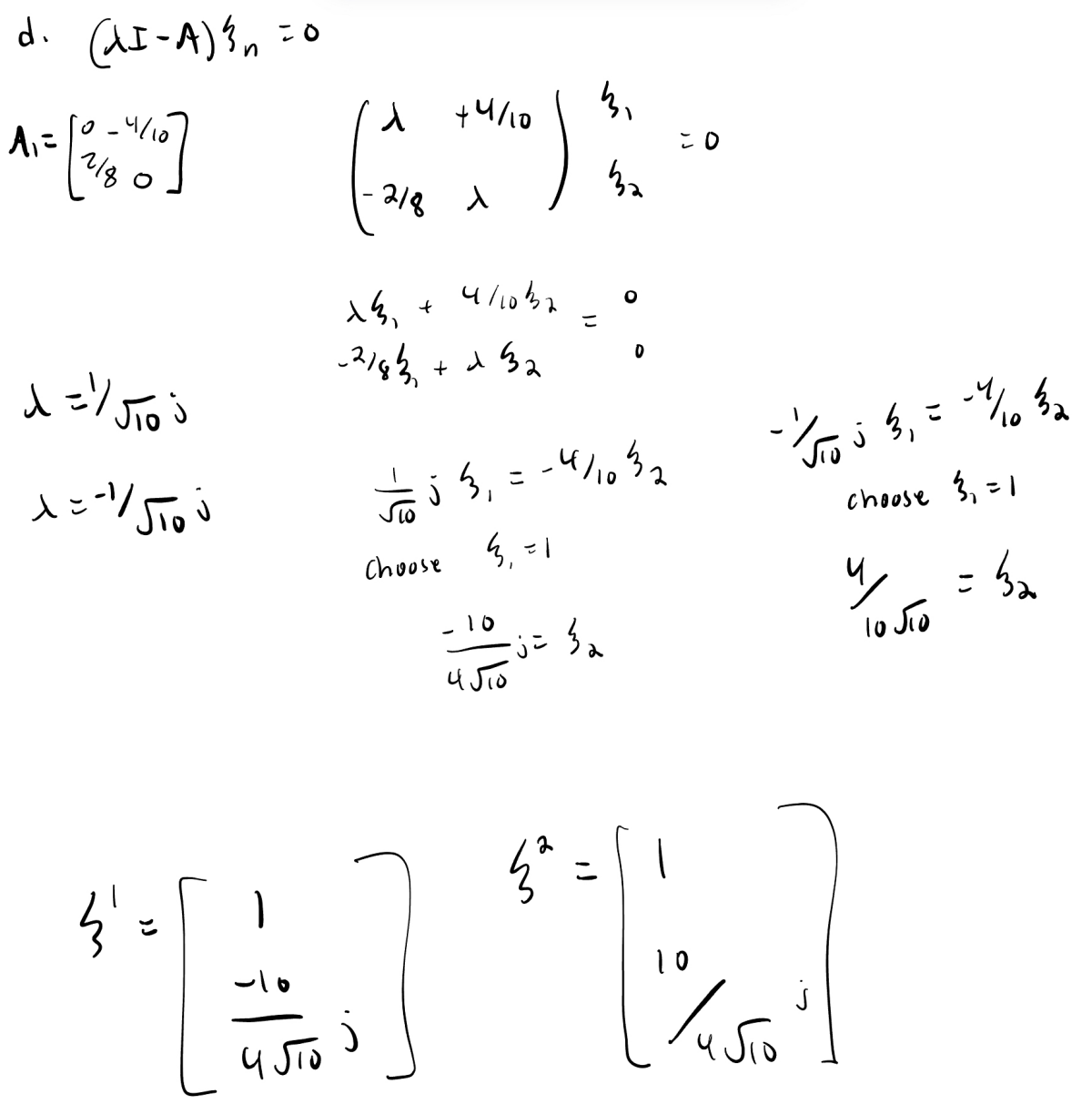


Figure 8: Eigenvector computation

Eigenvectors give directions in state-space. The change of basis to the eigenbasis scales and unscales by the magnitude of the eigenvectors (scalars pass through the change of basis $T^{-1}AT$).

e)

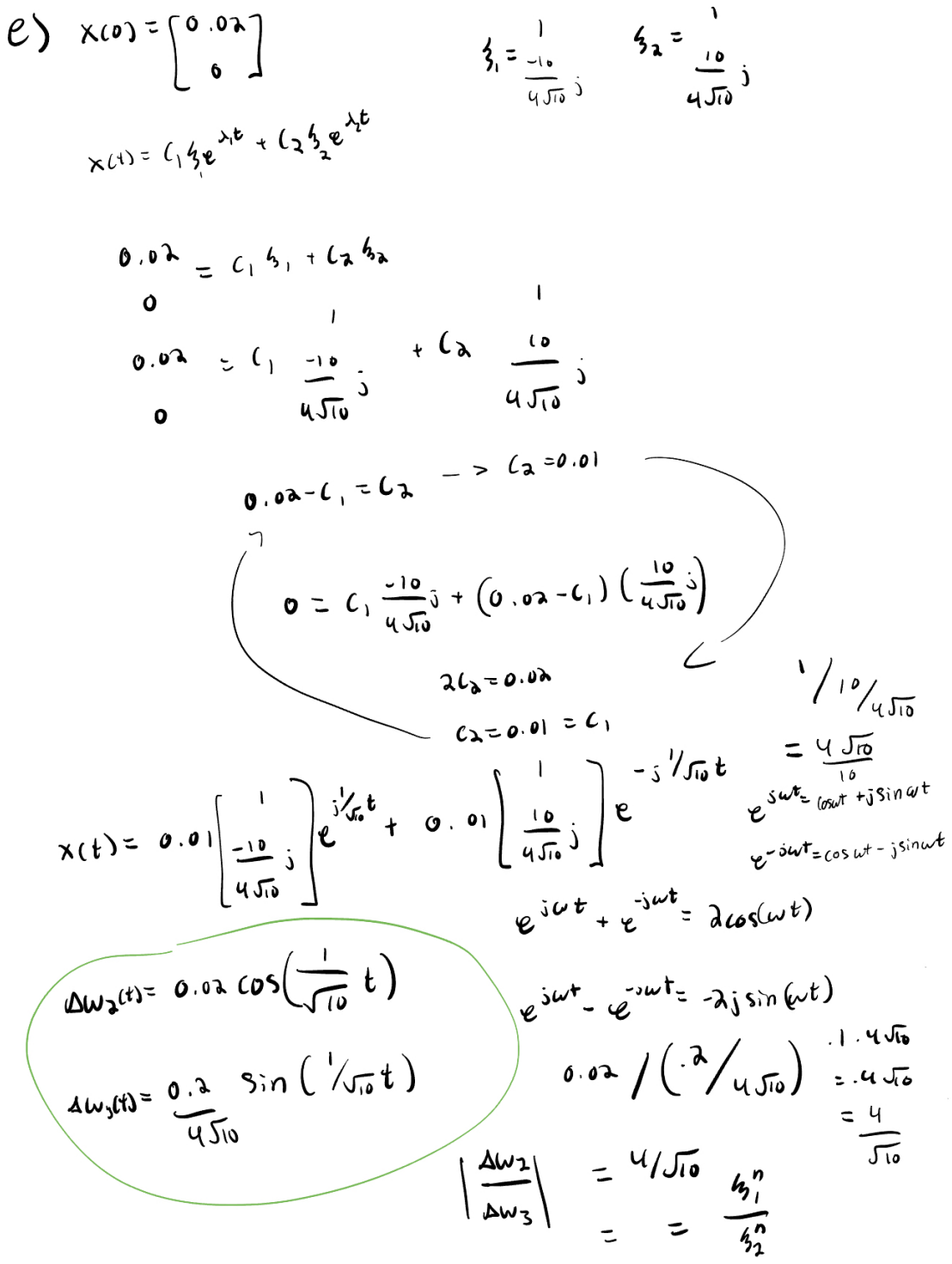


Figure 9: Time response

The ratio of their oscillation amplitudes and the ratio inferred from their eigenvalues is the same.

D

1.

a)

Verifications have been completed in the live scripts associated with this assignment: `https://github.com/antonioc76/vsc-assignment2`.

spin axis	eigenvalues	motion type	stability classification	relevant time-scale quantity
b_1	$\begin{bmatrix} 0.3162j, 0 \\ 0, -0.3162j \end{bmatrix}$	oscillatory	neutrally stable	$T = 19.87$
b_2	$\begin{bmatrix} 0.2041, 0 \\ 0, -0.2041 \end{bmatrix}$	exponential	unstable	$\tau = 4.9$
b_3	$\begin{bmatrix} 0.2582j, 0 \\ 0, -0.2582j \end{bmatrix}$	oscillatory	neutrally stable	$T = 24.33$

Table 1: Matlab linear system verifications

b)

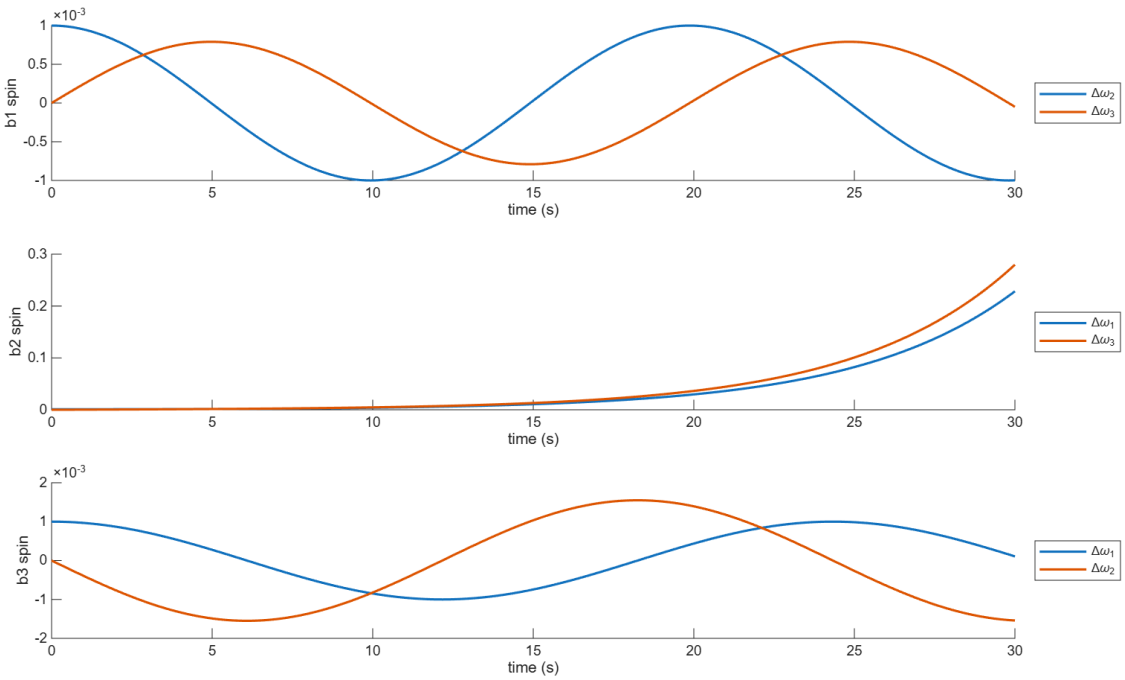


Figure 10: State space responses of linearized perturbation equations

c)

Rotation about the first and third axes is neutrally stable whereas rotation about the second axis is unstable. These results come from the ratios of sign of the 0th order term in the characteristic equation of each 2nd-order system, which is a function of the difference of the off axis moments of inertia.

E

1.

a)

False, the reference state may be any solution to the equations of motion. Linearization around a point, surface, or state space is based on a first order approximation of gradient at that point, which may be the zero vector, but is well defined at other points.

b)

False, a purely imaginary eigenvalue pair means that the the mode envelope does not grow or decay but the motion within that envelope is determined by the natural frequency. The natural frequency is the imaginary part in the case when the real part is zero, since the entire magnitude of the response is determined solely by the imaginary part.

c)

False, the pair represents one physical mode. The eigen directions are also different, so the mapping from the eigen space to the state space is different. The difference is exactly such that the the mode in state space is the same.

d)

False, they will decay to zero but not monotonically. Imagine a linear system with three negative real part eigenvalues, where only one is strictly real valued. The purely real valued mode will compose into state space as a monotonic decrease, but the imaginary mode from the complex eigen pair composes its oscillations into state space, resulting in (potentially ellipsoidal) oscillations.

F

1.

a)

The sorted principal moments of inertia are

$$I_1 = 162\ kgm^2 \tag{9}$$

$$I_2 = 108\ kgm^2 \tag{10}$$

$$I_3 = 72\ kgm^2 \tag{11}$$

The sorted normalized principal moments of inertia are

$$b_1 = \begin{bmatrix} -0.67 \\ -0.67 \\ -0.33 \end{bmatrix} \tag{12}$$

$$b_2 = \begin{bmatrix} 0.67 \\ 0.33 \\ -0.67 \end{bmatrix} \tag{13}$$

$$b_3 = \begin{bmatrix} -0.33 \\ 0.67 \\ -0.67 \end{bmatrix} \tag{14}$$

b)

Verified in the matlab script.

c)

The eigenvalues of the three single axis rotations are

spin axis	**eigenvalues**	**motion type**	**stability classification**
b_1	$\begin{bmatrix} 0.79j, 0 \\ 0, -0.79j \end{bmatrix}$	oscillatory	neutrally stable
b_2	$\begin{bmatrix} 0.4082, 0 \\ 0, -0.4082 \end{bmatrix}$	exponential	unstable
b_3	$\begin{bmatrix} 0.43j, 0 \\ 0, -0.43j \end{bmatrix}$	oscillatory	neutrally stable

Table 2: Torque free spin classifications

d)

The eigen pairs of the inertia tensor, comprised of the eigenvectors and eigenvalues, are the principal (independent) axes of rotation of the body and the resistance to torque in those directions respectively. The eigen pairs of the linearized dynamics matrix A on the other hand are are the directions (in principle axis coordinates) that rotational motion evolves when this body, as it rotates on one of these axes, experiences a small acceleration around another.